Turn Right

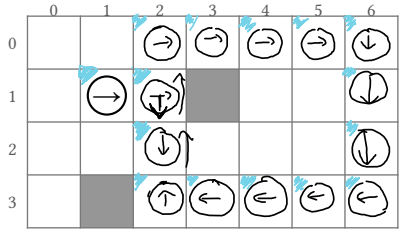
Name: _____ Date: _____

A robot vacuum cleans a room one square at a time. It knows one rule for what to do in one "step" of cleaning:

- If the square in front of it is open floor, it moves forward one square.
- If the square in front is a wall or furniture, it turns right a quarter turn and stays put.

Every square it leaves gets marked clean. Here it starts at (1, 1), facing right.

Q1. Be the robot. Draw what it will do for the first 10 steps. How many floor squares are not cleaned?



still dirty

Q2. What does a program need to remember about the room? About the robot?

Answer Box #1

Q3. What happens in one step?

Q4. When should the program stop?

Keeping a log

An `ArrayList` is a list that grows as you add to it.

```
record State(int x, int y, String facing) {}

ArrayList<State> log = new ArrayList<>();
log.add(new State(1, 1, "right"));
log.size()                              // 1
log.get(0)                              // State[x=1, y=1, facing=right]
log.contains(new State(1, 1, "right"))  // true
```

Imagine that before each step, the robot adds where it is and which way it faces to `log`.

Q5. Write the first six entries of `log` for the robot on the front.

Q6. How can `State` and the log help us decide when the robot should stop?

Q7. Why an `ArrayList` here, and not a `State[]`?

- Room width/length
- Robot location
- Clean squares
- Where obstacles are
- Robot facing

Turn that data into records/classes

Put at least 1 record/class in Answer Box #2

Questions:

- Represent room as 2D array or not?
- Represent obstacles + clean as part of same array, or separate list/other collection?
- Do any fields need to change? Does that imply classes?

2D array Room

Separate lists for obst. clean squares